Greecon
SH.P.K

02

HARDWARE SHEET HW02 · QUALIFIED BY SPECIFICATION

The converter (RS-485 to Ethernet)

The small unit that puts the equipment's data cable on the network, so that the box can reach every meter and module on it.

QUALIFIES	Qualified by specification. Any RS-485 to Ethernet converter that has a Modbus TCP to RTU gateway mode and takes the kit's serial settings and port 502.
USED IN	Kits H, S, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys, sets and fits the converter
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW02

The converter (RS-485 to Ethernet)

The meters, the modules and the field sensors speak Modbus RTU on a two-wire cable (RS-485). The box speaks Modbus TCP on the network. The converter sits between them: it puts the cable on the network, so that the box can reach every device on it. One of its settings matters more than all the others, and that is its mode.

At a glance

WHAT IT DOES	Puts the meters' cable (RS-485) on the network, so that the box can reach every meter and module on it. It must be in Modbus TCP to RTU gateway mode, and is set at the office (9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502).
HOW IT QUALIFIES	**Qualified by specification.** Any RS-485 to Ethernet converter that has a Modbus TCP to RTU gateway mode and takes the kit's serial settings and port 502.
SPECIFICATION	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode
FITTED BY	The project manager: set at the office, mounted and connected on site.
PLANNING ESTIMATE (EUR)	30 to 60 (basic) or 250 and up (industrial). Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. The box keeps one connection to a converter and asks one thing at a time, so any converter that does this will do.

It qualifies if:

- ☐ It converts RS-485 to Ethernet, fits a DIN rail and takes 9 to 36 V DC, so that it shares the 24 V supply.
- ☐ It has a **Modbus TCP to RTU gateway mode**. Makers also call it "Modbus gateway", "Modbus TCP server" or "protocol conversion".
- ☐ Its serial side can be set to 9600 baud, 8 data bits, no parity, 1 stop bit.
- ☐ Its TCP port can be set to 502.
- ☐ It can keep a fixed address on the network: set in the converter, or reserved for it in the router.

A basic converter costs about EUR 30 to 60 and an industrial one EUR 250 and up. Either qualifies.

STOP

Do not buy a converter that has only a “transparent” or “raw TCP” mode, or only “Modbus RTU over TCP”. In those modes the box’s requests arrive on the cable as noise, and nothing answers.

At each kind of site

Every site has one converter. What changes is what is behind it. The number after each device is its number on the cable.

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	In the enclosure. Behind it, on one cable: the grid meter (number 1), the solar meter (2), the water module (3) and the pump contact module (5).
S Solar plant with grid connection	IN THE KIT	1	In the enclosure. Behind it: the grid meter (1), the solar meter (2) and, if fitted, the load meter (4). A battery that speaks Modbus on its own network address does not need the converter; one on RS-485 joins the cable with a number of its own.
F Farm or greenhouse with irrigation	IN THE KIT	1	In the enclosure in the pump house. Behind it: the water module (3), the pump contact module (5) if there is a pump to watch, the soil probe (6; a second probe takes 8) and the weather sensor (7).
W Water tank and pump station	IN THE KIT	1	In the enclosure. Behind it: the water module (3), the pump contact module (5) and, if fitted, the pump meter (4).

Settings, and where they are made

Greecon sets the converter in its own web page at the office, before the kit leaves. On site you check its label. The settings are here in case a converter ever has to be set again.

Setting	Value
Mode	Modbus TCP to RTU (gateway)
Serial: speed	9600 baud, the same as every device on the cable
Serial: data bits, parity, stop bits	8, none, 1
TCP: port	502
Address on the network	A fixed one: a reservation for the converter in the customer’s router (on its MAC address), or an address set in the converter itself, on the same range as the router. Arranged on site, and written down

Write the settings on the converter’s label and photograph its settings page.

Fitting

PROJECT MANAGER

- [ ] Mount it on the DIN rail in the enclosure, on the low-voltage side, and power it from the 24 V supply.
- [ ] Connect the data cable: A to A, B to B, and the ground where there is one, on to the first device and from there to the next. The rules of the cable are on sheet HW03.
- [ ] Plug its network socket into the same router or switch as the box (or into the 4G router). Never a guest network.
- [ ] Check its label against this sheet: gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502.
- [ ] Ask the customer's network person for a fixed address for it, and write the address down. You do not type it anywhere: the box finds the converter.

In the platform

The converter is not a device in the platform. It is the address of every device behind it, shown on each as *Reached at 192.168.1.50:502, unit 3*.

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box tries port 502 on the private addresses of its own network (such as 192.168.x.x and 10.x.x.x), and behind whatever answers it asks unit numbers 1 to 10 a few read-only questions. It never writes to anything, and it takes about a minute.
- **If the converter is on another part of the network, or uses another port:** open **I know the converter's address**, type its address (each part without a leading zero: 192.168.1.50) and the port only if it is not 502, and press **Look at this address**. Only private addresses are accepted.
- **If its address changes later** (the router gave it a new one), every device behind it stops answering at the same moment. Press **Find equipment** again: the devices say **Found** at the new address, and **Use this address** sets them. Then ask for the fixed address.

Acceptance test

On a desk at the office, with the box:

- [ ] The converter's settings page shows gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit and port 502, and its label says the same.
- [ ] With one numbered device on its cable, the box finds the converter and that device at its own number:

```
greecon-edge find --host 192.168.1.50
```

It prints the converter's address and, under it, one line for each unit that answers, for example `unit 1: an Eastron meter, reading 1.80 kW at 231 V`.

- [ ] With two devices on the cable, each answers at its own number, with its own reading. If every number answers with the same words, the converter is not in gateway mode.

- [ ] A number that no device has does not answer.

On the day: Find equipment finds every device of the kit, and step 6, **Every device answers the box**, turns **Done** by itself.

What goes wrong

What you see	Likely cause	What to do
No Modbus converter answered on this network	The converter is off, or not on the same network as the box: a guest network, a second router, a cable in the wrong socket	Look at its power and network lights. Put the box and the converter on the same router. If it is on another part of the network, give its address
A converter answers at 192.168.1.50, but nothing answers behind it	The network side works and the cable side does not: A and B swapped, the converter not in gateway mode or not at 9600 baud, devices without power, a cable that is broken or branched	Swap the two wires at the converter. Check the mode and the speed in its settings page. Check that the devices have power
Units 1, 2 at 192.168.1.50 all answered with exactly the same words	The converter answers every unit number as the one device behind it: it is not in gateway mode	Set its mode to Modbus TCP to RTU (gateway), then press **Look again**
Every device behind it stops at the same moment (*no answer from 192.168.1.50:502 (unit 1)*), with the box online	The router gave the converter a new address	Press **Find equipment**, then **Use this address**. Ask for a fixed address, so that it does not happen again

More is in the runbooks (Project Manager Field Manual, PDF 02): R15 (a device is not found), R16 (a device is found at another address) and R17 (the converter's address changed).

Not yet proven

- This software has not yet looked at a real converter. It is tested against a pretend one that answers as the makers' documentation says. The first real kit is rehearsed on a desk before any site.
- Converters differ. Asked for a unit that is not there, some say so at once and some say nothing: the box handles both. One that accepts a single connection may drop the box's reading connection for a moment while the box looks, which costs one round of readings.
- A single device behind a converter in the wrong mode looks like a device at whichever number it is asked at: the box can tell the mode is wrong only when two units give the same answer. That is why the mode is checked on the settings page, and the first reading is compared with the meter's own display.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	